

Online Learning for Mean-Variance Portfolio Selection in Adversarial Environments with Side Information

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Abstract

Portfolio management relies heavily on Markowitz’s mean-variance optimization based on predictive signals about asset returns. However, this approach suffers from a lack of robustness during adversarial periods such as stock market crashes because predictions of returns and covariance become ineffective. Conversely, online learning algorithms offer performance guarantees in the worst cases, but rarely exploit predictive signals. This article seeks to reconcile these two approaches through the framework of Optimistic Online Convex Optimization. We introduce online portfolio selection algorithms that incorporate financial forecasts as hints to accelerate learning in calm periods while maintaining theoretical robustness in adversarial conditions. We establish new regret bounds that explicitly model the transition from a predictable regime to a market crisis. Empirically, our experiments on real market data demonstrate that optimistic strategies and meta-algorithms combining online learning and Markowitz portfolios outperform the Markowitz benchmark in terms of resilience to stock market crashes (Calmar ratio) and Sharpe ratio. Finally, we highlight the sensitivity of these approaches to the learning rate tuning, identifying a trade-off between theoretical performance and practical feasibility.

1 Introduction

Online Learning for portfolio selection presents a robust alternative to classical financial econometrics. Its primary advantage lies in its ability to operate within an adversarial setting, which allows for modeling without relying on restrictive statistical assumptions regarding assets return distribution. The vast majority of the literature in this field focuses on maximizing cumulative returns or the logarithm of cumulative wealth [1, 2, 3]. As detailed in the comprehensive survey [4], these works model portfolio selection as a repeated game over a horizon T between an agent (the investor) and an adversarial environment (the market). On each trading day $t = 1, \dots, T$:

1. At the beginning of the trading day t , the agent selects a portfolio vector w_t from the simplex $\Delta_N = \{w \in \mathbb{R}^N : w \succeq 0, \sum_{i=1}^N w_i = 1\}$, representing the allocation of wealth across N assets.
2. Subsequently, the market reveals the return vector $r_t \in \mathbb{R}^N$ (known at the end of the trading day).

The agent’s objective is to select w_t at the start of each trading period to maximize the (initially unknown) linear reward function $f_t(w_t) = \langle r_t, w_t \rangle$. To address this online linear optimization problem, many studies leverage algorithms from the "Follow-The-Regularized-Leader" (FTRL) family [5, 6]. Using a strictly convex regularization function $\mathcal{R}$ and learning rates $(\eta_t)_t$, standard FTRL computes the portfolio w_t as follows:

$$w_t = \arg \max_{w \in \Delta_N} \left(\sum_{s=1}^{t-1} \langle r_s, w \rangle - \frac{1}{\eta_t} \mathcal{R}(w) \right).$$

However, this theoretical framework faces two significant practical limitations. First, solely maximizing portfolio returns is insufficient as it fails to account for return variance (i.e., risk). To be viable for real-world applications, online portfolio selection methods should instead aim to maximize returns subject to a specific variance constraint. This approach aligns with the classical Markowitz Mean-Variance framework [7] and serves as a proxy for maximizing risk-adjusted metrics such as the realized Sharpe ratio, defined as $(w_t^\top r_t) / \sqrt{w_t^\top V_t w_t}$, where V_t represents the realized covariance matrix of asset returns. Second, this framework ignores available side information. In practice, quantitative investors generate forecasts for future returns and covariances—denoted as $\hat{r}_t$ and $\hat{V}_t$ —by applying machine learning and statistical techniques to diverse datasets. Despite being standard practice in modern asset management, current online portfolio selection frameworks rarely incorporate such predictions into their optimization process.

To bridge these two gaps, we formulate the portfolio selection problem within the framework of Optimistic Online Convex Optimization [8]. Unlike standard methods that assume total unpredictability, optimistic algorithms incorporate a 'hint'—in our case, the financial forecast of returns $\hat{r}_t$ and covariances $\hat{V}_t$ —into the update rule. This mechanism allows the algorithm to accelerate learning when predictions are accurate (small regret), while maintaining robust performance even if predictions turn out to be adversarial. Besides, we aim to maximize the risk-adjusted objective function:

$$f_t(w) = \langle r_t, w \rangle - \frac{\lambda}{2} w^\top V_t w \quad (1)$$

where λ is a risk aversion parameter. This formulation allows us to integrate volatility considerations in the portfolio construction process. The online learning approach also gives the advantage to be more robust to adversarial environments in which the forecasts $\hat{r}_t$ and $\hat{V}_t$ are performing poorly.

Contributions. Our main contributions can be summarized as follows:

- **Optimistic Algorithms for portfolios with volatility considerations:** We introduce a set of algorithms based on Optimistic Follow-The-Regularized-Leader (OFTRL) applied to the mean-variance objective (1). By treating market forecasts as "hints," these algorithms modulate their learning pace based on the quality of the predictions $\hat{r}_t$ and $\hat{V}_t$. Furthermore, we propose a Meta-Algorithm (Expert Aggregation) strategy for Online Ensemble Learning that dynamically allocates wealth between a classical Markowitz portfolio (pure exploitation of forecasts) and an online strategy (robust to adversarial environments), allowing for adaptive risk management.
- **Theoretical Guarantees:** We provide novel theoretical regret bounds for our algorithms under realistic market assumptions (Corollary 1 and 2), drawing inspiration from results of Optimistic Online Convex Optimization. The interest of these new regret bounds lies in their ability to model the performance of our algorithms during a market crash, specifically when transitioning from a period where return and covariance forecasts $\hat{r}_t$ and $\hat{V}_t$ are reliable to an adversarial, high-volatility phase where they fail. These bounds are also interesting because they are explicitly dependent on the accuracy of the return and covariance forecasts across both of these regimes.
- **Empirical Robustness Analysis:** We conduct experiments on real-world market data to evaluate the resilience of our framework. We specifically focus on adversarial environments—characterized by market crashes or high volatility regimes where traditional forecasts typically degrade. Using metrics such as Sharpe Ratio, Max Drawdown and Calmar Ratio, we show that our Online Learning approach significantly improves robustness compared to a standard Markowitz benchmark.

2 Algorithms

In this section, we detail the online learning algorithms employed. We first define the goal of the algorithms and the linearized loss framework, then introduce the Optimistic FTRL family (OGD and EG), and finally describe the adaptive learning rate mechanisms and the meta-algorithm used to aggregate experts.

2.1 Algorithms and regret minimization

The goal of the learner/investor on each trading day t is to choose a portfolio w_t that will maximize the reward function $f_t(w) = \langle w, r_t \rangle - \frac{\lambda}{2} w^\top V_t w$ based on the history of past portfolios and reward functions $w_1, f_1, \dots, w_{t-1}, f_{t-1}$ but without observing the actual function f_t . To solve this problem, we aim to develop decision-making algorithms—i.e., strategies for choosing a sequence of portfolios based on previously observed information—such that the cumulative reward $\sum_{t=1}^T f_t(w_t)$ is maximized. We measure the performance of an algorithm over T trading days through its regret R_T , defined as the difference between the cumulative reward yielded by the best portfolio over T trading days and the cumulative reward yielded by the sequence of portfolios $w_1, \dots, w_T$ that the learner chose:

$$R_T := \max_{w \in \Delta_N} \sum_{t=1}^T (f_t(w) - f_t(w_t)).$$

Another quantity that is often considered is the regret with respect to a fixed portfolio $w \in \Delta_N$, denoted $R_T(w)$:

$$R_T(w) := \sum_{t=1}^T (f_t(w) - f_t(w_t)).$$

2.2 Linearized Online Convex Optimization

Since the covariance term $w^\top V_t w$ in the function (1) is quadratic, We rely on the standard reduction from online convex optimization to online linear optimization: by concavity of f_t , we can write:

$$\forall w \in \Delta_N, f_t(w) - f_t(w_t) \leq \langle \nabla f_t(w_t), w - w_t \rangle$$

Summing over T and denoting $g_t = \nabla f_t(w_t) = r_t - \lambda V_t w_t$, we can see that the regret is bounded by the linearized regret:

$$\forall w \in \Delta_N, \sum_{t=1}^T (f_t(w) - f_t(w_t)) \leq \sum_{t=1}^T \langle g_t, w - w_t \rangle$$

Consequently, minimizing the regret with respect to the sequence of linear gradients g_t provides a valid upper bound on the regret of the original mean-variance objective. This approach allows us to avoid the computational cost of solving quadratic programs at every step while retaining theoretical guarantees. The algorithms described below utilize this gradient vector g_t (observed ex-post) and a prediction vector M_t (available ex-ante) to update the portfolio weights. The prediction M_t is constructed using the forecasted returns $\hat{r}_t$ and covariance $\hat{V}_t$:

$$M_t = \hat{r}_t - \lambda \hat{V}_t w_t$$

2.3 Optimistic Follow-The-Regularized-Leader (OFTRL)

To integrate predictive information, we employ the Optimistic Follow-The-Regularized-Leader (OFTRL) framework [8], [9, Sections 6.9 and 7.12]. Unlike standard FTRL, which regularizes around the previous portfolio w_t , Optimistic FTRL incorporates the "hint" M_t into the optimization step. The general update rule is:

$$w_{t+1} = \arg \max_{w \in \Delta_N} \left(\langle M_{t+1}, w \rangle + \sum_{s=1}^t \langle g_s, w \rangle - \frac{1}{\eta_{t+1}} \mathcal{R}(w) \right) \quad (2)$$

where $\mathcal{R}(w)$ is a strictly convex regularization function and η_t is the learning rate. We consider two specific instances of OFTRL based on the choice of the regularization function $\mathcal{R}(w)$:

Optimistic Online Gradient Descent (OGD): using the Euclidean regularization function $\mathcal{R}(w) = \frac{1}{2} \|w\|_2^2$, the update rule corresponds to a projected gradient descent step in the direction of the cumulative gradients plus the prediction.

$$w_{t+1} = \Pi_{\Delta_N} (w_t - \eta_{t+1} (g_t + M_{t+1})) \quad \text{where } \Pi_{\Delta_N}(\cdot) \text{ is the projection on the simplex.} \quad (3)$$

Optimistic Exponentiated Gradient (EG): using the Negative Entropic regularization function $\mathcal{R}(w) = -\sum_{i=1}^N w_i \ln w_i$, we obtain the Optimistic Exponentiated Gradient algorithm, which was notably studied in [10]. The update rule is:

$$w_{t+1} = \left(\frac{\exp(\eta_{t+1} G_{t,1})}{\sum_{j=1}^N \exp(\eta_{t+1} G_{t,j})}, \dots, \frac{\exp(\eta_{t+1} G_{t,N})}{\sum_{j=1}^N \exp(\eta_{t+1} G_{t,j})} \right) \quad \text{where } G_t = \sum_{s=1}^t (g_s + M_{s+1}) \quad (4)$$

2.4 Robustness via Adaptive Learning Rates

An important component of our approach is the choice of the learning rate η_t . We implement two adaptive strategies that will influence the algorithm's behavior in adversarial environments.

Gradient-Based Learning Rate: the standard adaptive approach scales the learning rate based on the historical volatility of the gradients, as recommended in [9, Section 7.6]:

$$\eta_{t+1} \propto \left(\sqrt{\sum_{s=1}^t \|g_s\|^2} \right)^{-1} \quad (5)$$

Prediction-Based Learning Rate: another approach is to use the adaptive rate proposed by [8], which depends on the prediction error:

$$\eta_{t+1} \propto \left(\sqrt{\sum_{s=1}^t \|g_s - M_s\|^2} \right)^{-1} \quad (6)$$

When the environment is easy and not adversarial, forecasts are good: $r_s \approx \hat{r}_s, V_s \approx \hat{V}_s$ so $g_s \approx M_s$. As a result, the denominator remains small, keeping η_t large. The algorithm learns aggressively and strongly relies on the hints, behaving nearly as well as the offline optimal strategy based on $\hat{r}_s$ and $\hat{V}_s$. On the contrary, in an adversarial environment in which forecasts are performing poorly, $g_s \gg M_s$. The learning rate η_t decreases quickly, forcing the algorithm to take smaller steps. This effectively prevents misleading forecasts of future returns and covariance matrix to cause large losses.

2.5 Meta-Algorithm for Online Ensemble Learning: aggregating Experts

To reconcile the potential of forecast-based Markowitz strategies with the worst-case protection of robust online learning, we introduce a hierarchical Meta-Algorithm based on the Prediction with Expert Advice framework [11]. Each algorithm/trading strategy is considered as an expert and the Meta-Algorithm assess each expert's performance based on their realized mean-variance reward $f_t(w_t^{\text{portfolio}})$.

Formulation: Consider the two following strategies/experts labelled 1, 2 respectively:

- $w_t^1 = \arg \max_{w \in \Delta_N} \langle \hat{r}_t, w \rangle - \frac{\lambda}{2} w^\top \hat{V}_t w$ is the Markowitz portfolio strategy based on forecasts $\hat{r}_t, \hat{V}_t$: this strategy exploits forecasts but is not robust to adversarial environments in which the forecasts perform poorly (e.g. when market volatility is high).
- $w_t^2 = \arg \max_{w \in \Delta_N} \left(\sum_{s=1}^t \langle g_s, w \rangle - \frac{1}{\eta_{t+1}} \mathcal{R}(w) \right)$ is the (non-optimistic) FTRL portfolio strategy. It does not exploits forecasts but is robust to adversarial environments.

On each trading day t , the meta-algorithm maintains a probability distribution $q_t \in \Delta_2$ over these two strategies. The aggregated portfolio w_t^{meta} is the weighted average of the experts' portfolios:

$$w_t^{\text{meta}} = q_{t,1} w_t^1 + q_{t,2} w_t^2$$

The weights are updated according to the exponential weights rule.

$$\forall k \in \{1, 2\}, \quad q_{t+1,k} = \frac{q_{t,k} \cdot \exp(\eta_{\text{meta}} f_t(w_t^k))}{q_{t,1} \cdot \exp(\eta_{\text{meta}} f_t(w_t^1)) + q_{t,2} \cdot \exp(\eta_{\text{meta}} f_t(w_t^2))}$$

where η_{meta} is the learning rate of the Meta-algorithm. This formulation allows to dynamically switch from the Markowitz forecast-based portfolio to the adversarially robust FTRL portfolio:

- In non adversarial environments: forecasts of returns and covariance perform well, so the forecast-based Markowitz expert generates high returns with controlled variance, $f_t(w_t^1)$ is high, and the meta-algorithm concentrates capital on this portfolio.
- In adversarial environments (crashes and periods of high volatility): the forecasts perform poorly, $f_t(w_t^1)$ decreases. Due to the exponential update, Markowitz portfolio's weight $q_{t,1}$ is aggressively penalized. The capital then flows toward the more adversarially robust Online Learning expert.

3 Theoretical guarantees

3.1 General results on OFTRL

Let us first introduce the following theorem from [9]:

Theorem 1 (Regret Bound for Adaptive OFTRL). *Let $K \subseteq \mathbb{R}^d$ be a non-empty closed convex set. Consider the problem of Online Convex Optimization where the learner aims to maximize a sequence of concave reward functions $f_t : K \rightarrow \mathbb{R}$. Let $\mathcal{R} : K \rightarrow \mathbb{R}$ be a 1-strongly convex regularization function with respect to a norm $\|\cdot\|$. The Optimistic Follow-The-Regularized-Leader (OFTRL) algorithm update is:*

$$w_{t+1} = \arg \max_{w \in K} \left(\sum_{s=1}^t \langle g_s, w \rangle + \langle M_{t+1}, w \rangle - \frac{\mathcal{R}(w)}{\eta_t} \right)$$

where $g_t \in \partial f_t(w_t)$, M_t is a predictive hint for the gradient g_t , and $(\eta_t)_{t \geq 1}$ is a non-increasing sequence of positive learning rates. Then, for any fixed comparator $u \in K$, the regret $R_T(u)$ is bounded by:

$$R_T(u) = \sum_{t=1}^T (f_t(u) - f_t(w_t)) \leq \frac{\mathcal{R}(u)}{\eta_{T+1}} - \frac{\mathcal{R}(w_1)}{\eta_1} + \sum_{t=1}^T \left(\frac{\eta_t}{2} \|g_t - M_t\|_*^2 + \frac{\mathcal{R}(w_{t+1})}{\eta_t} - \frac{\mathcal{R}(w_{t+1})}{\eta_{t+1}} \right)$$

where $\|\cdot\|_*$ denotes the dual norm of $\|\cdot\|$.

The proof of this theorem can be found in [9, Section 7.12] (it is called theorem 7.36 in this book).

3.2 Regret Bounds under Regime Shift

In this section, we present new theoretical regret bounds and we analyze the behavior of our Optimistic FTRL algorithm in a "regime shift" scenario:

Assumption 1 (Regime switch). Denote $\|\cdot\|_2$ the $\mathbb{R}^N$ ℓ_2 norm and $\|\cdot\|_{2,S}$ the $\mathbb{R}^{N \times N}$ ℓ_2 matrix norm. Let us assume there exists $t_0, \epsilon_1, \epsilon_2, C_1, C_2$ with $C_1 \gg \epsilon_1, C_2 \gg \epsilon_2$, such that:

1. $\forall 1 \leq t \leq t_0, \|r_t - \hat{r}_t\|_2^2 \leq \epsilon_1$ and $\|V_t - \hat{V}_t\|_{2,S}^2 \leq \epsilon_2$ (Accurate Regime).
2. $\forall t_0 < t \leq T, \|r_t - \hat{r}_t\|_2^2 \leq C_1$ and $\|V_t - \hat{V}_t\|_{2,S}^2 \leq C_2$ (Adversarial Regime).

This hypothesis assumes the occurrence of a market crash or a high-volatility regime starting at t_0 during the investment period $[1, T]$. Before t_0 , covariance and return forecasts are accurate because statistical regularities exist in the assets' returns: we are in a benign environment and we can exploit these regularities to build predictors of returns and covariance. After t_0 , high volatility occurs, these regularities vanish, and forecasts become terrible: we are now in a highly adversarial environment.

Corollary 1 (Robustness of the Optimistic OGD portfolio strategy to regime switch). Denote $(w_t)_{t=1}^T$ the sequence of portfolios chosen by Optimistic OGD with update rule given by (3) and learning rate $\eta_t = 1/\left(2\sqrt{\sigma + \sum_{s=1}^{t-1} \|g_s - M_s\|_2^2}\right) \forall t$, for some $\sigma > 0$. Let assumption 1 hold. If we set $\sigma \geq C_1 + \lambda^2 C_2$, the regret of Optimistic OGD satisfies:

$$R_T = \max_{w \in \Delta_N} \sum_{t=1}^T (f_t(w) - f_t(w_t)) \leq 1.8\sqrt{\sigma + t_0(\epsilon_1 + \lambda^2 \epsilon_2) + (T - t_0)(C_1 + \lambda^2 C_2)}$$

Moreover, if we know C_1, C_2 we can set $\sigma = C_1 + \lambda^2 C_2$ and we get:

$$R_T = \mathcal{O}\left(\sqrt{\epsilon_1 + \lambda^2 \epsilon_2} \sqrt{t_0}\right) + \mathcal{O}\left(\sqrt{C_1 + \lambda^2 C_2} \sqrt{T - t_0 + 1}\right)$$

The proof of this corollary can be found in appendix.

Remarks:

1. If forecasts of returns and covariance are perfect during the accurate regime, ie if $\epsilon_1, \epsilon_2 \rightarrow 0$, we get $R_T = \mathcal{O}\left(\sqrt{C_1 + \lambda^2 C_2} \sqrt{T - t_0 + 1}\right)$.
2. Standard (Non-Optimistic) Online Gradient Descent achieves a regret bound of $\mathcal{O}(G\sqrt{T})$, where G is such that $\|r_t - \lambda V_t w_t\| \leq G \forall t$. In financial terms, G is related to the maximum market volatility. Even if the market is perfectly predictable up to t_0 , standard OGD—which ignores predictive hints—would still incur a regret of $\mathcal{O}(G\sqrt{t_0})$ during the first phase. Hence, as long as ϵ_1, ϵ_2 are small enough, we get an improvement over the Standard OGD, while maintaining the same robust $\mathcal{O}(\sqrt{T})$ guarantees as standard OGD in worst-case scenarios in which $t_0 = 1$ and the environment is not predictable at all.

We get similar bounds for the Optimistic EG portfolio strategy but with less restrictive assumptions regarding the regime switch since it now depends on the ℓ_∞ norm.

Assumption 2 (Regime switch). Denote $\|\cdot\|_\infty$ the $\mathbb{R}^N$ ℓ_∞ norm and $\|\cdot\|_{\infty,S}$ the $\mathbb{R}^{N \times N}$ ℓ_∞ matrix norm. Let us assume there exists $t_0, \epsilon_1, \epsilon_2, C_1, C_2$ with $C_1 \gg \epsilon_1, C_2 \gg \epsilon_2$, such that:

1. $\forall 1 \leq t \leq t_0, \|r_t - \hat{r}_t\|_\infty^2 \leq \epsilon_1$ and $\|V_t - \hat{V}_t\|_{\infty,S}^2 \leq \epsilon_2$ (Accurate Regime).
2. $\forall t_0 < t \leq T, \|r_t - \hat{r}_t\|_\infty^2 \leq C_1$ and $\|V_t - \hat{V}_t\|_{\infty,S}^2 \leq C_2$ (Adversarial Regime).

Corollary 2 (Robustness of the Optimistic EG portfolio strategy to regime switch). Denote $(w_t)_{t=1}^T$ the sequence of portfolios chosen by Optimistic EG with update rule given by (4) and learning rate $\eta_t = \ln(N)/\left(\sqrt{\sigma + \sum_{s=1}^{t-1} \|g_s - M_s\|_\infty^2}\right) \forall t$, for some $\sigma > 0$. Let assumption 1 hold. If we set $\sigma \geq C_1 + \lambda^2 C_2$, the regret of Optimistic EG satisfies:

$$R_T = \max_{w \in \Delta_N} \sum_{t=1}^T (f_t(w) - f_t(w_t)) \leq (\sqrt{2} \ln(N) + 1) \sqrt{\sigma + t_0(\epsilon_1 + \lambda^2 \epsilon_2) + (T - t_0)(C_1 + \lambda^2 C_2)}$$

Moreover, if we know C_1, C_2 we can set $\sigma = C_1 + \lambda^2 C_2$ and we get:

$$R_T = \mathcal{O}\left((\sqrt{2\ln(N)} + 1)\sqrt{\epsilon_1 + \lambda^2 \epsilon_2} \sqrt{t_0}\right) + \mathcal{O}\left((\sqrt{2\ln(N)} + 1)\sqrt{C_1 + \lambda^2 C_2} \sqrt{T - t_0 + 1}\right)$$

4 Numerical experiments

4.1 Dataset

In this section, we use a dataset of historical returns and intraday realized volatility for major stock market indices (e.g., Euro Stoxx 50, S&P 500), sourced from the Oxford-Man Institute's Realised Library. The objective of this experiment is to implement the algorithms presented in Section 2 and to construct daily-updated portfolios based on $N = 22$ indices¹ and their intraday realized returns and covariances². The realized covariance between two assets i and j on day t (intraday covariance) is estimated from the realized intraday volatility as follows:

$$Cov_{t,i,j} = \rho_{t,i,j} \sigma_{t,i} \sigma_{t,j}$$

where $\sigma_{t,i}$ is the realized volatility of index i during day t (originally computed in the dataset using 5-minute frequency data points and initially available in the dataset), and $\rho_{t,i,j}$ is the historical correlation between the returns of indices i and j , calculated over a 60-day rolling window up to time t .

4.2 Predictors and models

In our experiments, we employ straightforward predictors for future daily returns and covariances, defined as the rolling mean of daily realized returns and covariances over the trailing one-year period (typically consisting of 252 trading days):

$$\hat{r}_t = \frac{1}{252} \sum_{i=1}^{252} r_{t-i} \quad \text{and} \quad \hat{V}_t = \frac{1}{252} \sum_{i=1}^{252} V_{t-i}$$

The interest of these simple predictors is that during low-volatility regimes, asset returns often exhibit temporal persistence and identifiable statistical patterns (e.g., momentum). In such environments, historical averages serve as reliable proxies for future behavior, so $\hat{r}_t$ and $\hat{V}_t$ are effective predictors. Conversely, during market crashes or high-volatility episodes (i.e., adversarial periods), these statistical regularities tend to break down, causing the predictive power of historical averages to deteriorate as returns lose their trend-following characteristics.

We implement the algorithms that we presented in section 2. The name of each algorithm, as it appears in the plots, is indicated in parentheses:

1. **Online strategies that do not rely on forecasts:** This category includes OGD (OGD Standard) and EG (EG Standard), both utilizing an adaptive learning rate based solely on past gradients g_t , with the shape (7).
2. **Online strategies that rely on forecasts:** This category includes Optimistic OGD and Optimistic EG. For each algorithm, we implement two variants: one with an adaptive learning rate based on past gradients g_t with the shape (7) (OGD Opt (Grad LR) and EG Opt (Grad LR)), and another based on both past gradients g_t and past predictions M_t with the shape (8) (OGD Opt (Pred LR) and EG Opt (Pred LR)). Additionally, we include the Markowitz portfolio, computed daily using return and covariance forecasts (Markowitz Rolling).
3. **Ensemble online learning strategies:** These consist of Meta-Algorithms aggregating previous strategies. We consider: a Meta-Algorithm based on all aforementioned strategies (Meta: All Experts); a Meta-Algorithm combining the forecast-based Markowitz portfolio with non-optimistic OGD (Meta: Marko + OGD Std); and a Meta-Algorithm combining the forecast-based Markowitz portfolio with non-optimistic EG (Meta: Marko + EG Std).

¹It should be noted that indices cannot be traded directly; ETFs must be used to build portfolios tracking these indices.

²Due to missing price and intraday volatility data in the dataset, indices with more than 10% of missing values were removed from the analysis, so we end up with 22 indices in the dataset.

4.3 Practical remarks on the implementation

A limitation of the adaptive learning rate chosen as (5) and (6) is that the learning rate decreases monotonically regardless of prediction quality. Even if forecasts are perfect, the algorithm slows down, failing to fully exploit the easiness of the environment. To prevent this, in practice, we introduce a decay parameter γ set to $\gamma = 0.9$ and we write:

$$\eta_{t+1} \propto 1/\sqrt{\gamma \sum_{s=1}^{t-1} \|g_s\|^2 + (1-\gamma)\|g_t\|^2} \quad \text{for the learning rate based solely on gradients;} \quad (7)$$

$$\eta_{t+1} \propto 1/\sqrt{\gamma \sum_{s=1}^{t-1} \|g_s - M_s\|^2 + (1-\gamma)\|g_t - M_t\|^2} \quad \text{for the learning rate based on gradients and predictions.} \quad (8)$$

A limitation of the optimistic formula update (2) is that when the number of trading days t becomes very large, the portfolio chosen by Optimistic OGD/EG converges to the portfolio chosen by OGD/EG. This is caused by the fact that the reliance on prediction M_{t+1} vanishes through time. When t is very large:

$$\sum_{s=1}^t g_s + M_{t+1} \approx \sum_{s=1}^t g_s$$

To solve this problem, we multiply the prediction M_{t+1} by t so that it is at the same scale at the cumulative sum of gradients, which yields $\sum_{s=1}^t g_s + tM_{t+1} \approx \sum_{s=1}^t g_s$.

4.4 Results of the experiments

We begin by conducting a sensitivity analysis of the various online algorithms to evaluate how their annualized realized Sharpe Ratio evolves as a function of the learning rate (specifically, the numerator parameter used in formulas (7) and (8)). The Annualized Realized Sharpe Ratio is defined as $SR_{\text{ann}} = \sqrt{252} \cdot \frac{\hat{\mu}}{\hat{\sigma}}$, where $\hat{\mu}$ is the empirical mean of the daily portfolio returns and $\hat{\sigma}$ is their empirical standard deviation. It is the reference metric to look at the realized risk-adjusted return of a strategy as mentioned in section 1.

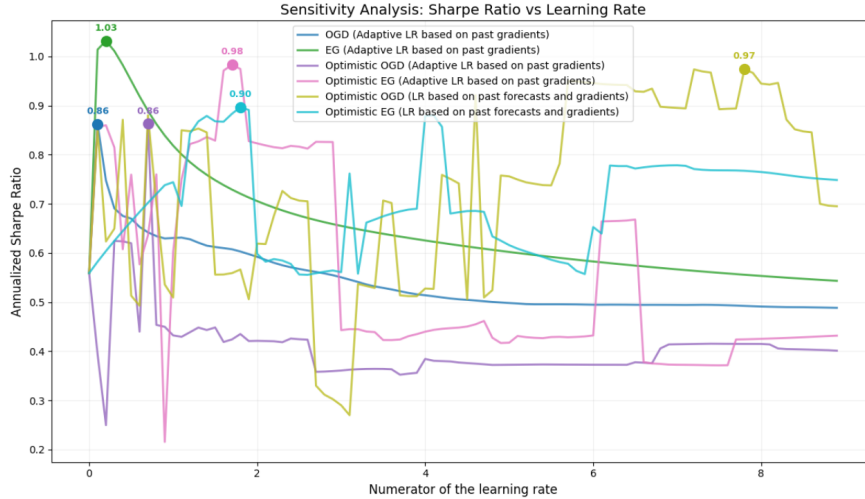


Figure 1: Sensitivity Analysis: Annualized Sharpe Ratio vs Learning Rate.

Figure 1 illustrates the very strong sensitivity of optimistic strategies to learning rate tuning, where a marginal increment of 10^{-1} in the learning rate's numerator can trigger large fluctuations in the Sharpe Ratio (up to the order of 10^{-1}). This instability represents a practical limitation compared to standard non-optimistic OGD and EG, which exhibit much smoother and more predictable performance profiles.

Subsequently, we select the optimal learning rate for each strategy—defined as the one maximizing the Sharpe Ratio.

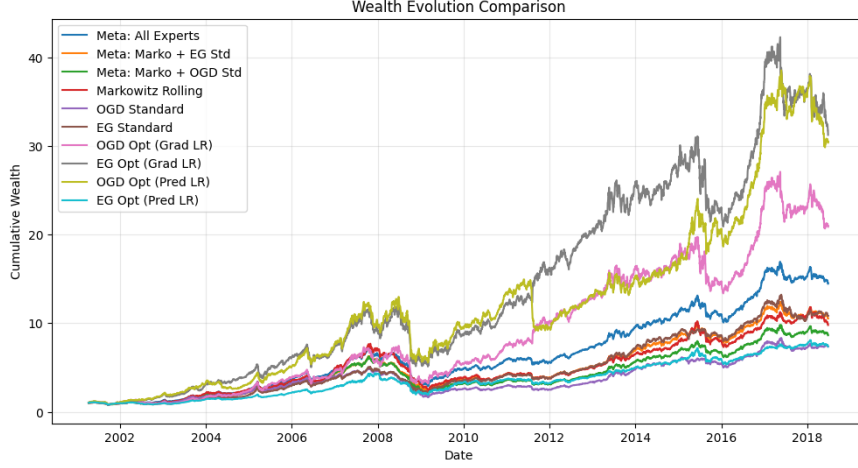


Figure 2: Evolution of Cumulative wealth of each strategy.

Figure 2 illustrates the cumulative wealth generated by each strategy over the 2002-2018 period. We can see the significant dominance of specific optimistic variants. The Optimistic EG with learning rate based on gradients (gray line) and Optimistic OGD with learning rate based on predictions (olive line) drastically outperform the market, achieving a final wealth up to 3 times higher than the Markowitz strategy (red line). Moreover, the results highlight a strong disparity within the optimistic family. For instance, the Optimistic EG with gradients based on predictions (cyan line) fails to outperform the Markowitz strategy.

Finally, we compare the performance of the strategies using three key metrics: the Sharpe Ratio, the Maximum Drawdown (MDD), and the Calmar Ratio. The Maximum Drawdown measures the largest peak-to-trough decline in portfolio wealth W_t over the period $[0, T]$:

$$\text{MDD} = \min_{t \in [0, T]} \left(\frac{W_t}{\max_{s \in [0, t]} W_s} - 1 \right)$$

The Calmar Ratio compares the annualized return $\hat{\mu}$ to this maximum risk: $\text{Calmar Ratio} = \frac{\hat{\mu}}{|\text{MDD}|}$

These metrics are particularly well-suited for assessing robustness in adversarial environments. While the Sharpe Ratio penalizes both upside and downside volatility symmetrically, the Maximum Drawdown specifically quantifies the worst-case loss during market crashes (adversarial regimes). Consequently, the Calmar Ratio is an ideal measure of return adjusted to tail risk (risk of large losses) and allows to identify most robust strategies to high-volatility adversarial periods.

Strategy	Sharpe Ratio	Max Drawdown	Calmar Ratio
Meta: All Experts	1.10	-59.72%	0.27
Meta: Marko + EG Std	1.05	-63.79%	0.22
Meta: Marko + OGD Std	0.95	-67.49%	0.19
OGD Standard	0.86	-67.21%	0.18
EG Standard	1.03	-58.63%	0.24
OGD Opt (Grad LR)	0.86	-57.26%	0.34
EG Opt (Grad LR)	0.97	-57.51%	0.38
OGD Opt (Pred LR)	0.97	-57.19%	0.38
EG Opt (Pred LR)	0.90	-56.11%	0.22
Markowitz (Benchmark)	0.95	-68.32%	0.20

Table 1: Comparison of Risk-Adjusted Performance Metrics.

Table 1 details the risk-adjusted performance of the implemented strategies. Two major trends emerge from the analysis of Sharpe Ratios, Maximum Drawdowns, and Calmar Ratios.

First, regarding robustness to market crashes, the Optimistic strategies demonstrate a clear superiority over the classical Markowitz benchmark. While the Markowitz portfolio suffers a catastrophic Maximum Drawdown of -68.32% , the Optimistic variants (notably EG Opt and OGD Opt with Gradient Learning Rates) successfully limit losses to the -57% range. This resilience is best captured by the Calmar Ratio,

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6 Appendix: supplementary material

Proof of Corollary 1 and 2. Let us give a more general proof that works for any strongly convex regularizer wrt a norm $\|\cdot\|$. Denote $D = \sup_{a,b \in \Delta_N} \mathcal{R}(a) - \mathcal{R}(b)$ and let us consider the following learning rate:

$$\eta_{t+1} = \frac{D}{\left(\sqrt{\sigma + \sum_{s=1}^t \|g_s - M_s\|_*^2}\right)}$$

Denote $\|\cdot\|_*$ the dual $\mathbb{R}^N$ norm to $\|\cdot\|$ and $\|\cdot\|_{*,S}$ the $\mathbb{R}^{N \times N}$ matrix norm associated. Let us assume there is a change of regime during the trading period, ie there exists $t_0, \epsilon_1, \epsilon_2, C_1, C_2$ with $C_1 \gg \epsilon_1, C_2 \gg \epsilon_2$, such that:

1. $\forall 1 \leq t \leq t_0, \|r_t - \hat{r}_t\|_*^2 \leq \epsilon_1$ and $\|V_t - \hat{V}_t\|_{*,S}^2 \leq \epsilon_2$ (Accurate Regime).
2. $\forall t_0 < t \leq T, \|r_t - \hat{r}_t\|_*^2 \leq C_1$ and $\|V_t - \hat{V}_t\|_{*,S}^2 \leq C_2$ (Adversarial Regime).

By theorem 1, for any $u \in \Delta_N$:

$$\begin{aligned} R_T(u) &\leq \frac{\mathcal{R}(u) - \mathcal{R}(w_1)}{\eta_{T+1}} + \sum_{t=1}^T \frac{\eta_t}{2} \|g_t - M_t\|_*^2 \text{ for any sequence of decreasing learning rates } (\eta_t)_t. \\ &\leq \frac{D}{\eta_{T+1}} + \sum_{t=1}^T \frac{\eta_t}{2} \|g_t - M_t\|_*^2 \\ &\leq \sqrt{\sigma + \sum_{t=1}^T \|g_t - M_t\|_*^2} + \sum_{t=1}^T \frac{\eta_t}{2} \|g_t - M_t\|_*^2 \text{ by definition of } \eta_{T+1} \end{aligned}$$

Moreover, for any positive sequence $(x_t)_{t=1}^T$, for any $\sigma \geq \max_{1 \leq t \leq T} x_t$, we have the following inequalities (see e.g. [9, Lemma 4.13]):

$$\begin{aligned} \frac{\sqrt{\sigma + \sum_{s=1}^t x_s}}{\sqrt{\sigma + \sum_{s=1}^{t-1} x_s}} &\leq 2 \\ \sum_{t=1}^T \frac{x_t}{\sum_{s=1}^t x_s} &\leq 2 \sqrt{\sum_{t=1}^T x_t} \end{aligned}$$

Applying these inequalities yields:

$$\begin{aligned} R_T(u) &\leq \sqrt{\sigma + \sum_{t=1}^T \|g_t - M_t\|_*^2} + D \sqrt{2} \sqrt{\sum_{t=1}^T \|g_t - M_t\|_*^2} \\ &\leq (\sqrt{2}D + 1) \sqrt{\sigma + \sum_{t=1}^T \|g_t - M_t\|_*^2} \\ &\leq (\sqrt{2}D + 1) \sqrt{\sigma + t_0(\epsilon_1 + \lambda^2 \epsilon_2) + (T - t_0)(C_1 + \lambda^2 C_2)} \\ &\leq (\sqrt{2}D + 1) \sqrt{t_0(\epsilon_1 + \lambda^2 \epsilon_2) + \sqrt{(T - t_0)(C_1 + \lambda^2 C_2) + \sigma}} \text{ since for } a, b > 0, \sqrt{a+b} \leq \sqrt{a} + \sqrt{b} \end{aligned}$$

Where the antepenultimate inequality comes from the fact that $\|g_t - M_t\|_*^2 = \|r_t - \hat{r}_t + \lambda(V_t - \hat{V}_t)w_t\|_*^2 \leq \|r_t - \hat{r}_t\|_*^2 + \lambda^2 \|V_t - \hat{V}_t\|_{*,S}^2$ and then applying the change of regime assumption.

The result follows by noticing that $D = \frac{1}{2}$ and $\|\cdot\| = \|\cdot\|_2$ when the regularizer is $\mathcal{R}(w) = \frac{1}{2} \|w\|_2^2$ (OGD case) and $D = \ln N$ and $\|\cdot\| = \|\cdot\|_1$ when the regularizer is the negative entropy (EG case). $\square$